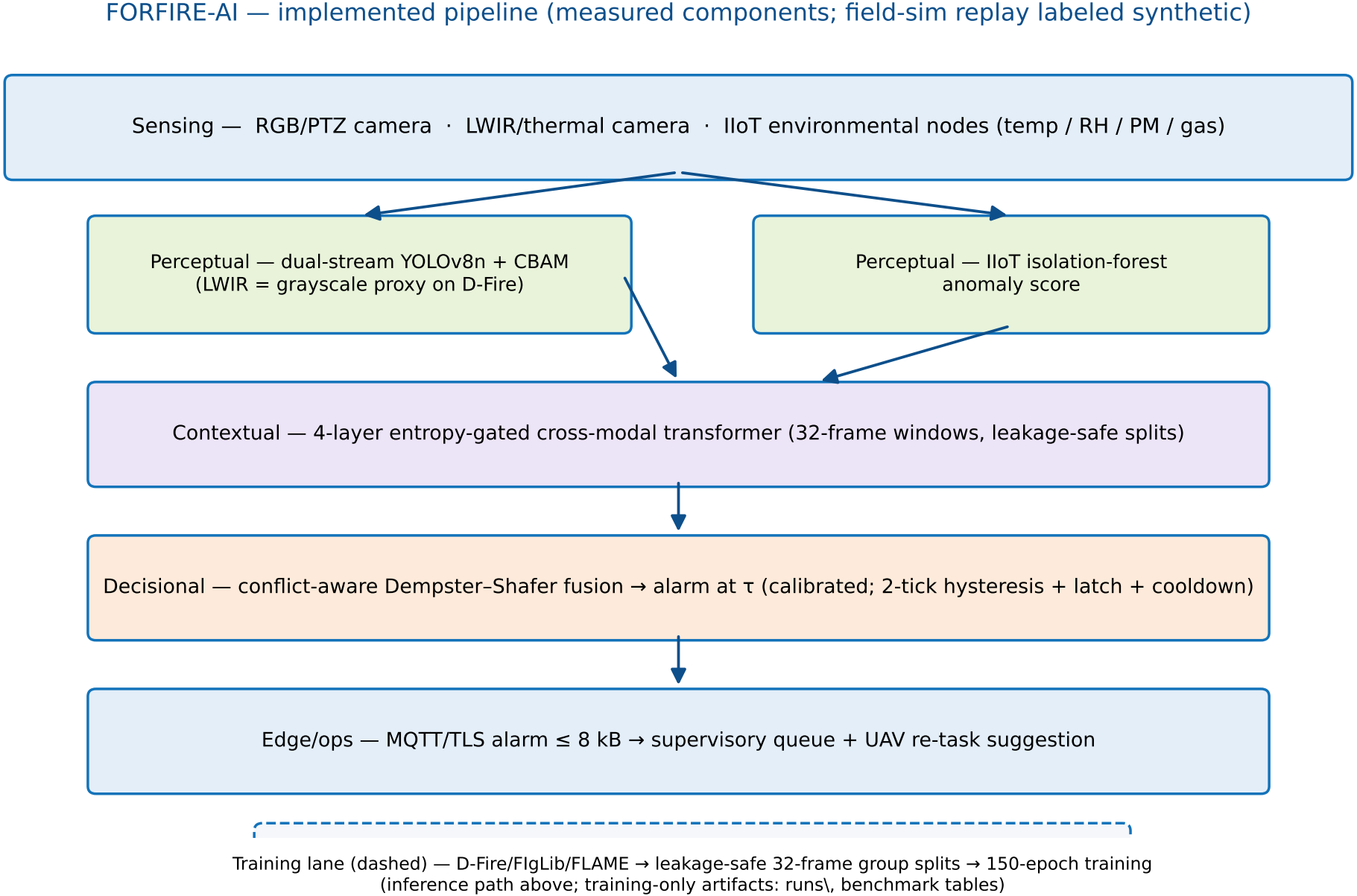
FORFIRE-AI — implemented pipeline (measured components; field-sim replay labeled synthetic)
Sensing — RGB/PTZ camera · LWIR/thermal camera · IIoT environmental nodes (temp / RH / PM / gas)
Perceptual — dual-stream YOLOv8n + CBAM
(LWIR = grayscale proxy on D-Fire)
Perceptual — IIoT isolation-forest
anomaly score
Contextual — 4-layer entropy-gated cross-modal transformer (32-frame windows, leakage-safe splits)
Decisional — conflict-aware Dempster–Shafer fusion → alarm at τ (calibrated; 2-tick hysteresis + latch + cooldown)
Edge/ops — MQTT/TLS alarm ≤ 8 kB → supervisory queue + UAV re-task suggestion
Training lane (dashed) — D-Fire/FIgLib/FLAME → leakage-safe 32-frame group splits → 150-epoch training
(inference path above; training-only artifacts: runs\, benchmark tables)